



COMPONENTS

System-on-module

iWave Systems has launched a system-on-module (SOM) based on Xilinx's Zynq UltraScale+ MPSoC. The SOM delivers exceptional results in rugged environments and offers outstanding performance, flexibility and security in such applications as



advanced driver assistance systems, cloud computing security, machine learning, artificial intelligence (AI) and data center virtualisation.

It incorporates 4x Arm Cortex A53 application processor at 1.5GHz, 2x Cortex R5 real-time processor at 600MHz, ARM Mali 400 MP2 GPU, memory interfaces (4GB DDR4 FPGA memory and 8GB eMMC flash) and up to one million programmable logic cells. It supports a rich variety of connectivity peripherals such as PCIe, USB3.0, SATA 3.1, display port and Gigabit Ethernet through GTR transceivers.

iWave Systems

www.iwavesystems.com

Microcontrollers

The new STM32H7 microcontrollers (MCUs) from STMicroelectronics (ST) are the industry's highest-performing Arm Cortex-M general-purpose MCUs, combining dual-core punch



with power-saving features and enhanced cyber protection. The new devices leverage a 480MHz version of Cortex-M7, the highest performing member of Arm's Cortex-M family, and add a 240MHz Cortex-M4 core.

With ST's smart architecture, efficient L1 cache and adaptive real-time ART Accelerator, the MCUs set new speed records at 1327 DMIPS and 3224 CoreMark executing from embedded

flash. ST's Chrom-ART Accelerator delivers an extra boost to graphics performance. To maximise energy efficiency, each core operates in its own power domain and can be turned off individually when not needed.

STMicroelectronics India

www.st.com

Analogue frontend

Analog Devices has introduced AD4110-1 analogue frontend with integrated 24-bit ADC for industrial process control systems. AD4110-1 is a universal input AFE with an integrated ADC that allows customers to design a platform input module that they can configure for multiple functions. This saves significant R&D cost, reduces time-to-market and requires less design resources.

The high voltage inputs of AD4110-1 are fully software configurable for current or voltage signals, allowing direct interface to all standard industrial analogue signal sources. One reference design replaces many, reducing module size and decreasing cost of ownership. Software-configurable I/O is a key factor in the implementation of Industry 4.0.

Analog Devices Inc.

www.analog.com

TVS diode arrays

Littelfuse has introduced a series of low-capacitance TVS diode arrays (SPA diodes) optimised to protect high-speed differential data lines from electrostatic discharge (ESD), cable discharge events (CDE), electrical fast transients (EFT) and lightning-induced surges, keeping network communications reliable by safeguarding signal integrity.



The SP3384NUTG series can protect up to four channels against up to 15A (IEC 61000-4-5 second edition) and

up to $\pm 30\text{kV}$ ESD (IEC 61000-4-2), and is available in a compact μDFN package. The combination of low capacitance and low clamping voltage make SP3384NUTG a robust protection solution for 2.5G/5G/10G Ethernet high-speed data interfaces without signal degradation, resulting in high reliability for a variety of applications.

Littelfuse Inc.

www.littelfuse.com

32-bit microcontrollers

Renesas Electronics has unveiled Renesas Advanced (RA) Family of 32-bit Arm Cortex-M microcontrollers (MCUs). The MCUs deliver the ultimate combination of optimised performance, security, connectivity, peripheral IP and easy-to-use Flexible Software Package (FSP) to address the next generation of embedded solutions. To support the new family, Renesas has built a comprehensive partner ecosystem to deliver an array of software and hardware building blocks that will work out of the box with the MCUs.



Renesas Electronics Corp.

www.renesas.com

Baseband processors

NXP Semiconductors N.V. has announced a new series of Layer-scape Access processors, designed for 5G Access Edge applications. Layer-scape Access processors target a variety of deployment scenarios compliant with O-RAN Alliance specifications, including central units/distributed units (CU/DU), radio units (RUs), along with integrated small cells and customer premises equipment (CPE) systems.

NXP Semiconductors N.V.

www.nxp.com

Small form factor system-on-module

Apalis iMX8 is a small form factor system-on-module based on the latest